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The Editorial Office
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Dear Editor,

We submit for your consideration the manuscript “**Contact without Commitment: Atomistic Characterisation of β -Lactoglobulin Adsorption Dynamics at the Air–Water Interface**” by Chalakon Pornjariyawatch and Prapasiri Pongprayoon (Department of Chemistry, Faculty of Science, Kasetsart University, Bangkok, Thailand).

β -Lactoglobulin (BLG), the dominant whey protein in bovine milk, is a central determinant of dairy foam stability, yet the molecular mechanism by which it adsorbs at the air–water interface has remained uncharacterised at atomic resolution. We address this gap with the first unbiased atomistic molecular dynamics study of native-state BLG at the air–water interface, accumulating **4.00 μ s** of trajectory across one bulk-start simulation and three near-interface replicas using the CHARMM36m force field and a $12 \times 12 \times 35$ nm slab geometry.

Our principal findings are: (i) BLG contacts the interface frequently — **613 discrete events** across 4.00 μ s, resolved at single-atom contact distance — yet 607 of these events terminate without commitment, revealing a genuine kinetic bottleneck rather than a sampling limitation; (ii) calyx SASA (24–37 nm², corrected for periodic-boundary-condition artefacts) and orientation are **statistically independent** ($r = +0.006$; block bootstrap 95% CI [−0.09, +0.11]), ruling out simple two-factor coupling as the commitment trigger; (iii) the protein remains **globally compact** throughout 4 μ s of aggregate trajectory, inconsistent with pre-adsorption global unfolding models; and (iv) **Loop CD/EF** (residues 57–60) emerges as the dominant calyx-mobility region near the interface, replacing Loop BC in the bulk. These results establish a precise atomistic baseline for the pre-commitment contact ensemble — the missing reference state for designing enhanced-sampling calculations of the adsorption pathway.

We believe this work is well-suited for *JCIS* given the journal’s strong coverage of protein behaviour at fluid interfaces, foam and emulsion stabilisation mechanisms, and colloid science at the molecular scale. The findings are directly relevant to the design of functional dairy ingredients and to the broader question of how globular proteins cross the air–water interface.

The manuscript has not been submitted elsewhere and is not under consideration at any other journal. All authors have approved the submission. The authors declare no competing financial interests.

The dataset supporting this study has been deposited at Zenodo (DOI: [10.5281/zenodo.XXXXXX](https://doi.org/10.5281/zenodo.XXXXXX)), and all analysis scripts are available at github.com/cpornjar/blg-airwater-interface-md.

We are grateful for your consideration and look forward to your response.

Sincerely,

Chalakon Pornjariyawatch
PhD Student, COMFHA Laboratory

Department of Chemistry, Faculty of Science
Kasetsart University, Bangkok 10900, Thailand
chalakon.p@ku.th

Prapasiri Pongprayoon (corresponding author)

Associate Professor, COMFHA Laboratory
Department of Chemistry, Faculty of Science
Kasetsart University, Bangkok 10900, Thailand